

FFT-Based Spectral Analysis and Chatter Detection across Varying Depths of Cut at Constant Spindle Speed

1. Introduction

Machining is one of the most widely used manufacturing processes for producing components with high dimensional accuracy and surface quality. During machining operations, cutting forces generated at the tool–workpiece interface play a significant role in determining machining performance, tool life, dimensional accuracy, and surface finish. Variations in machining parameters, particularly the depth of cut (DOC), directly influence these cutting forces and may lead to unstable cutting conditions.

One of the most critical forms of machining instability is **chatter, a self-excited vibration** that arises from the dynamic interaction among the cutting tool, workpiece, and machine structure. Chatter results in poor surface finish, accelerated tool wear, increased machining noise, reduced dimensional accuracy, and, in severe cases, damage to the machine tool. Therefore, early identification and analysis of chatter are essential for selecting stable machining parameters and improving manufacturing productivity.

The dynamic behavior of a machining process can be effectively studied through cutting force signals acquired during machining. These signals contain valuable information regarding the interaction between the cutting tool and the workpiece and can be analyzed in both the time and frequency domains. Time-domain statistical features such as mean, root-mean-square (RMS), standard deviation, peak-to-peak value, crest factor, skewness, and kurtosis provide information about the amplitude characteristics and statistical distribution of the force signals. Frequency-domain analysis using the Fast Fourier Transform (FFT) reveals dominant vibration frequencies and periodic components associated with the cutting process, while Power Spectral Density (PSD) estimation using Welch's method provides a more robust representation of the signal's energy distribution across the frequency spectrum.

In this study, machining force data corresponding to different depths of cut are analyzed to investigate the influence of cutting conditions on process stability. The recorded force signals are first preprocessed by identifying the active cutting region to eliminate idle machine motion. Subsequently, statistical features are extracted in the time domain, frequency characteristics are obtained through FFT, and PSD analysis is computed using Welch's method. In addition, Pearson correlation analysis is employed to evaluate the relationship between different force components, and an empirical power-law model is developed to describe the variation of cutting force with depth of cut. Finally, a chatter index based on normalized force variability is used to classify the stability of each machining condition.

The overall objective of this work is to develop a systematic methodology for analyzing machining force signals and extracting meaningful indicators of machining performance and

process stability. The study demonstrates how signal-processing techniques can be applied to machining data to characterize cutting behavior under varying depths of cut and provide quantitative measures that support machining process evaluation.

2. Objective

The objectives of this study are:

- To analyze machining force signals obtained under different depths of cut (DOC).
- To extract and evaluate time-domain statistical features of the cutting force signals.
- To perform frequency-domain analysis using FFT and Power Spectral Density (PSD).
- To investigate the relationship between cutting forces and depth of cut.
- To assess machining stability using a force-based chatter index.
- To compare the machining conditions through graphical and statistical analysis.

3. Experimental Data

The experimental dataset consists of cutting force measurements acquired during turning operations under various combinations of spindle speed and depth of cut (DOC). Each dataset contains the time history of the three orthogonal cutting force components:

- **Feed Force (F_x) (N)**
- **Radial Force (F_y) (N)**
- **Cutting Force (F_z) (N)**

The experiments were performed at multiple spindle speeds ranging from **52 RPM to 550 RPM** and depths of cut ranging from **100 μm to 2600 μm** . The available machining conditions are summarized in the following table.

Table 3.1 Available Experimental Machining Conditions

Spindle Speed (RPM)	Available Depth of Cut, DOC (μm)
52	800, 1600, 2600
114	200, 600, 800, 1000, 1200, 1400, 1800, 2200
148	400, 1000, 1200, 1400, 1600, 1800, 2000, 2200
192	200, 400, 800, 1000, 1200
250	200, 400, 600, 800, 1000
325	200, 400, 600, 700, 800
420	100, 200, 400, 600, 700, 800, 900, 1200, 1400
550	200, 400, 600, 800, 1000, 1100

The recorded force signals were stored in CSV format and subsequently used for preprocessing, time-domain feature extraction, frequency-domain analysis using the Fast Fourier Transform (FFT), Power Spectral Density (PSD) estimation, correlation analysis, and machining stability assessment.

4. Methodology

4.1 Data Preprocessing

The raw cutting-force data acquired during the turning experiments were initially recorded in **Technical Data Management Streaming (TDMS)** format using the data acquisition system. For subsequent processing and analysis, the TDMS files were converted into **Comma-Separated Values (CSV)** format, providing a convenient structure for data handling and feature extraction.

The recorded force signals inevitably contain segments that do not correspond to actual material removal, such as tool approach, air cutting before tool engagement, and tool retraction after the machining pass. If these non-cutting regions are retained, the calculated statistical features, including mean force, root-mean-square (RMS), and standard deviation, become biased toward the idle machining condition rather than the actual cutting process. Consequently, preprocessing is performed to isolate the portion of the signal corresponding to steady-state cutting.

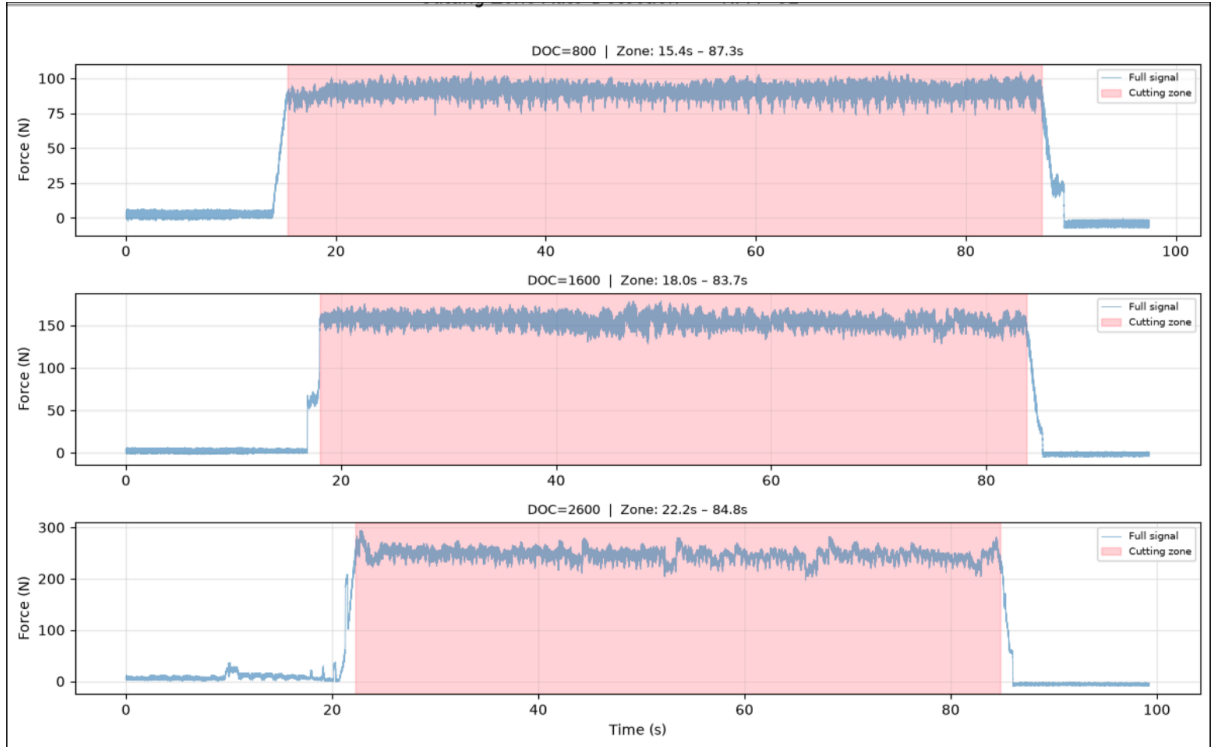
As an initial step, incomplete or invalid data records are removed to ensure data integrity. In addition, the initial transient samples, which may contain sensor settling effects and machine start-up vibrations, are excluded from further analysis. These preprocessing operations improve the reliability and consistency of the extracted features and ensure that the subsequent analyses accurately represent the machining process.

4.2 Cutting Zone Identification

Since the recorded force signals contain both cutting and non-cutting regions, it is essential to identify the interval corresponding to actual material removal before extracting features. During machining, the cutting force increases significantly once the cutting tool engages the workpiece and remains relatively high throughout the cutting operation. This characteristic provides an effective basis for identifying the active cutting region.

In this work, an amplitude-based thresholding technique is employed to isolate the cutting zone. A threshold is determined from the statistical distribution of the force signal, and the continuous portion of the signal exceeding this threshold is identified as the active cutting interval. Because the threshold is derived directly from the measured force data, the method automatically adapts to different spindle speeds and depths of cut without requiring external trigger signals or manual adjustment for individual experiments.

The extracted cutting interval is subsequently used for all time- and frequency-domain analyses, ensuring that the computed features represent only the steady-state machining process and enabling consistent comparisons across different experimental conditions.



4.3 Time-Domain Statistical Features

Time-domain features are the simplest class of vibration/force-signal descriptors: they summarize a signal's statistical shape directly, without any transformation, and are widely used in condition monitoring because they are computationally cheap and physically interpretable.

Let (x_i) denote the (i^{th}) force sample and let N be the total number of samples in the extracted cutting interval.

Mean Force:

The mean value represents the average level of the cutting force during the selected cutting interval:

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$$

Root Mean Square (RMS):

RMS represents the overall energy or magnitude of the force signal and is particularly useful for comparing the severity of cutting conditions at different depths of cut.

$$x_{RMS} = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2}$$

Standard Deviation:

It describes the variability of the cutting force around its mean value. An increase in standard deviation may indicate greater fluctuations in the cutting process.

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2}$$

Peak-to-Peak Value:

This parameter represents the total amplitude range of the cutting-force signal and is useful for identifying large fluctuations in force.

$$x_{P-P} = x_{max} - x_{min}$$

Kurtosis:

Kurtosis is used to characterize the shape of the probability distribution of the force signal:

$$K = \frac{1}{N} \sum_{i=1}^N \left(\frac{x_i - \bar{x}}{\sigma} \right)^4$$

A higher kurtosis value indicates a distribution with more pronounced extreme deviations or impulsive events.

Skewness:

The skewness of the force signal is calculated as

$$S = \frac{1}{N} \sum_{i=1}^N \left(\frac{x_i - \bar{x}}{\sigma} \right)^3$$

Skewness indicates the degree of asymmetry in the distribution of forces around their mean.

Dimensionless signal indicators:

In addition to the conventional statistical parameters, three dimensionless indicators are calculated.

The **crest factor** is

$$CF = \frac{x_{peak}}{x_{RMS}}$$

where x_{peak} is the maximum absolute value of the force signal.

The **shape factor** is defined as

$$SF = \frac{x_{RMS}}{\frac{1}{N} \sum_{i=1}^N |x_i|}$$

and the **impulse factor** is

$$IF = \frac{x_{peak}}{\frac{1}{N} \sum_{i=1}^N |x_i|}$$

These dimensionless quantities provide additional information about the shape and variability of the cutting-force signal and can be useful for identifying changes in machining behavior.

4.4 Frequency-Domain Analysis Using FFT

After extracting the steady-state cutting interval, frequency-domain analysis is performed using the Fast Fourier Transform (FFT). While time-domain parameters describe the magnitude and statistical characteristics of the force signal, the FFT allows the signal's periodic components to be identified by their frequencies. The study uses FFT to identify dominant vibration frequencies and frequency components associated with the machining process.

For a discrete force signal $x[n]$, the discrete Fourier transform is

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}$$

where N is the number of samples and k represents the frequency index. The corresponding frequency is

$$f_k = \frac{k}{N\Delta t}$$

Where Δt is the sampling interval?

In the present analysis, a **Hanning window** is applied to the cutting-force signal before FFT computation. Windowing reduces spectral leakage and improves the identification of dominant frequency components.

The spindle frequency is calculated from the spindle speed as

$$f_s = \frac{RPM}{60}$$

The corresponding spindle harmonics are

$$f_n = nf_s, \quad n = 1, 2, 3, \dots$$

In the implemented analysis, the first six spindle harmonics are considered.

The following frequency-domain parameters are extracted:

- Dominant frequency
- Spectral centroid
- Spectral bandwidth
- Harmonic energy ratio

Dominant frequency:

The dominant frequency is the frequency corresponding to the maximum magnitude of the FFT spectrum:

$$f_{dom} = f_k \text{ for } k = \arg \max |X[k]|$$

Spectral centroid:

The spectral centroid is calculated as

$$f_c = \frac{\sum_{k=1}^N f_k |X[k]|^2}{\sum_{k=1}^N |X[k]|^2}$$

and represents the center of gravity of the frequency spectrum.

Spectral bandwidth:

The spectral bandwidth is calculated as

$$B = \sqrt{\frac{\sum_{k=1}^N (f_k - f_c)^2 |X[k]|^2}{\sum_{k=1}^N |X[k]|^2}}$$

It represents the spread of spectral energy around the spectral centroid.

Harmonic Energy Ratio:

While the spectral centroid and bandwidth describe the overall distribution of spectral energy, they do not directly indicate how much of the signal energy is concentrated around the expected spindle-related frequency components. Therefore, the harmonic energy ratio is calculated.

The harmonic energy ratio is defined as:

$$HR = \frac{\sum_{n=1}^6 \sum_{f_k \in [f_n - BW, f_n + BW]} |X[k]|^2}{\sum_{k=1}^N |X[k]|^2}$$

Where:

- HR = harmonic energy ratio
- $X[k]$ = FFT of the cutting-force signal
- f_s = spindle frequency
- n = spindle harmonic number
- BW = bandwidth around each harmonic

In the present analysis:

$BW = 2 \text{ Hz}$ and $n = 1, 2, \dots, 6$ are used.

A higher HR value indicates that a larger proportion of the FFT energy is concentrated around the spindle harmonics. In contrast, a lower value indicates that more energy occurs outside these expected harmonic bands.

4.5 Power Spectral Density Analysis

To obtain a more robust representation of the frequency distribution of the cutting-force signal, Power Spectral Density (PSD) analysis is performed using Welch's method. PSD analysis provides an estimate of how the signal power is distributed over frequency and is less sensitive to individual spectral fluctuations than a single FFT estimate. The methodology specified for the study includes Welch PSD analysis as part of the frequency-domain investigation.

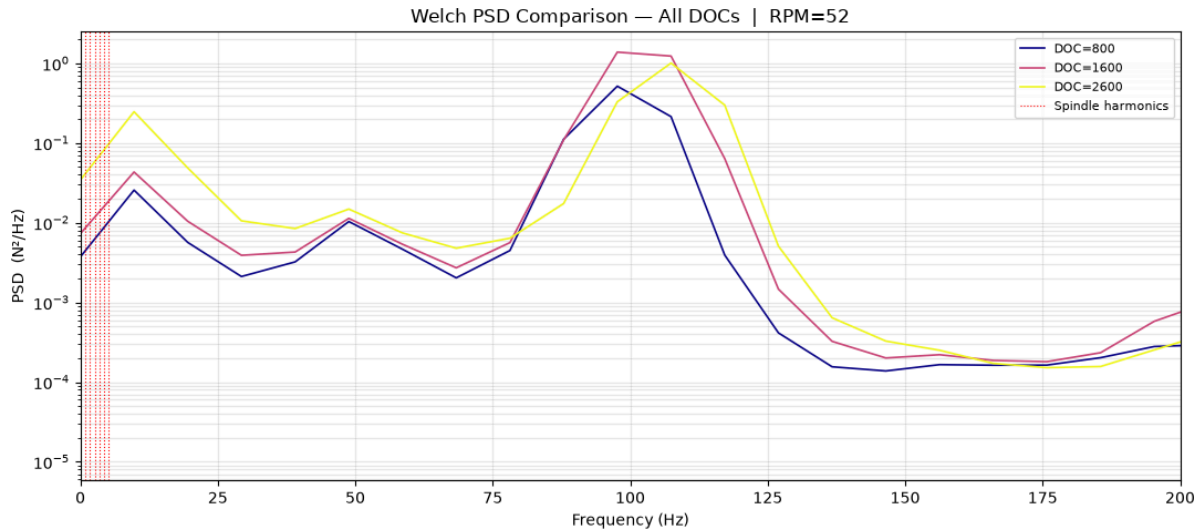
For a force signal, the PSD can be expressed in its basic periodogram form as:

$$PSD(f) = \frac{1}{N f_s} |X(f)|^2$$

where $PSD(f)$ is the power spectral density at frequency f , $X(f)$ is the Fourier transform of the force signal, N is the number of samples, and f_s is the sampling frequency.

The Welch method divides the signal into overlapping segments, calculates the periodogram of each segment, and averages the resulting spectra. This reduces the variance of the spectral estimate and provides a smoother representation of the frequency content.

The PSD is particularly important for the present study because it is subsequently used to quantify the proportion of force-signal energy occurring around the spindle frequency and its harmonics.



The figure illustrates a Welch Power Spectral Density (PSD) comparison across three Depths of Cut (DOC = 800, 1600, and 2600) at a constant spindle speed of 52 RPM. The frequency response highlights a dominant structural resonance band between 90 Hz and 115 Hz. As the depth of cut increases, broadband vibration energy increases significantly, and the primary peak shifts to the right due to changes in dynamic tool stiffness. While the low-frequency spikes below 10 Hz represent stable, forced spindle harmonics, the massive energy amplification near 100 Hz tracks the system's progression toward instability. By mapping these energy thresholds, the PSD analysis successfully predicts active tool chatter. The results confirm that while the system remains stable at lower cutting parameters (DOC = 800), increasing the depth of cut to 1600 and 2600 triggers a clear transition into fully active, regenerative tool chatter.

4.6 Chatter Index and Stability Classification

A force-based chatter index is developed to quantify the deviation of the cutting-force spectrum from the expected spindle-related frequency components.

For the n^{th} spindle harmonic, a frequency band of width $\pm BW$ is considered:

$$f_n = n f_s$$

$$f_n - BW \leq f \leq f_n + BW$$

where, in the present implementation,

$$BW = 2 \text{ Hz}$$

and the first six spindle harmonics are considered.

Let E_{total} represent the total PSD energy, and E_{spindle} represent the energy contained within the selected spindle-harmonic bands. The chatter index is defined as

$$CI = 1 - \frac{E_{\text{spindle}}}{E_{\text{Total}}}$$

with the value restricted to the range

$$0 \leq CI \leq 1$$

In the present study, the total spectral energy and the energy contained within the spindle-harmonic bands are obtained from the **Welch Power Spectral Density (PSD)** of the cutting-force signal. The PSD is calculated using overlapping windowed segments and averaged to obtain a smoother spectral estimate. The energy within each selected harmonic band is then summed to obtain E_{spindle} .

The chatter index, therefore, differs from the harmonic energy ratio introduced in Section 4.4. The harmonic energy ratio is calculated from the Hanning-windowed FFT magnitude, whereas the chatter index is calculated from the Welch PSD. The chatter index is used as the primary frequency-domain indicator for the subsequent stability classification.

Thus, a low CI value indicates that a larger fraction of the signal energy is concentrated around the spindle harmonics. In contrast, a high value indicates that a larger fraction of energy lies outside these bands.

In the present implementation, the stability boundary is determined separately for each spindle speed using the midpoint between the minimum and maximum chatter-index values:

$$CI_{threshold} = \frac{CI_{min} + CI_{max}}{2}$$

The machining condition is classified as **Stable** when

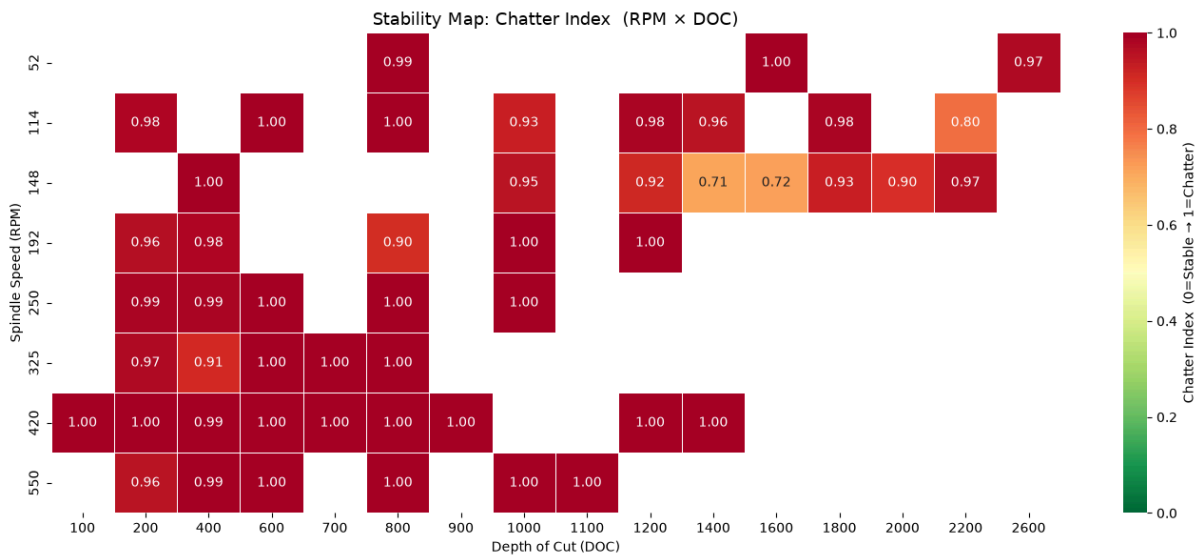
$$CI \leq CI_{threshold}$$

and as **Chatter** when

$$CI > CI_{threshold}$$

This classification should be regarded as **empirical, based on the available force data**, rather than an independently validated physical chatter criterion.

4.6.1 Stability Map



To visualize the variation in machining stability across the investigated operating conditions, the chatter index is plotted as a function of spindle speed and depth of cut. A stability map is generated by plotting spindle speed against depth of cut and representing the corresponding chatter index using a color scale. This allows the variation of frequency-based stability behavior across different combinations of spindle speed and depth of cut to be examined.

4.7 Effect of Depth of Cut on Cutting Force

The influence of depth of cut on cutting-force behavior is investigated by comparing extracted statistical parameters across different DOC values.

The primary quantities considered are:

- Mean cutting force
- RMS cutting force

- Standard deviation
- Peak-to-peak force
- Dominant frequency
- Chatter index

The variation of these parameters with DOC is plotted to identify systematic changes in the machining process. Increasing DOC is expected to modify the cutting load and may also influence the dynamic stability of the machining operation.

The relationship between mean cutting force and DOC is additionally represented using an empirical power-law model:

$$F = C (DOC)^n$$

where F is the mean cutting force, C is an empirical cutting-force coefficient, and n is the fitted exponent.

The parameters C and n are obtained by nonlinear curve fitting using the experimental data. The coefficient of determination R² is subsequently used to evaluate the goodness of fit.

4.8 Pearson Correlation Analysis

To investigate the relationship between depth of cut and the extracted force and frequency-domain features, Pearson correlation analysis is performed. In addition, pairwise correlations among the extracted features are evaluated to identify relationships and potential redundancy among the indicators.

$$r_{xy} = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^N (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^N (y_i - \bar{y})^2}}$$

Where:

- r_{xy} = Pearson correlation coefficient
- x_i = i^{th} observation of variable x
- y_i = i^{th} observation of variable y
- $\bar{x}$ = Mean of variable x
- $\bar{y}$ = Mean of variable y
- N = Number of observations

4.9 Time-Frequency Analysis Using Spectrogram

The FFT and Welch PSD analyses provide an overall representation of the frequency content of the selected cutting interval. However, these approaches do not directly show how the frequency content changes with time. To investigate time-varying behavior within the cutting process, a Short-Time Fourier Transform (STFT)- based spectrogram analysis is performed.

$$X(t, f) = \sum_n x[n]w[n - t]e^{-j2\pi fn}$$

Where:

X(t,f) = Short-Time Fourier Transform at time t and frequency f

$x[n]$ = cutting-force signal
 $w[n-t]$ = window function centered around time t
 n = discrete time index

In the present implementation, the spectrogram is calculated using segments of:

$$N_{\text{segment}} = 256 \text{ samples}$$

with an overlap of:

$$N_{\text{overlap}} = 200 \text{ samples}$$

The spectrogram power is represented in decibels as:

$$P_{\text{dB}} = 10 \log_{10}(S_{xx} + 10^{-12})$$

where S_{xx} represents the estimated power spectral density at each time-frequency point, the small numerical term 10^{-12} is included to avoid taking the logarithm of zero.

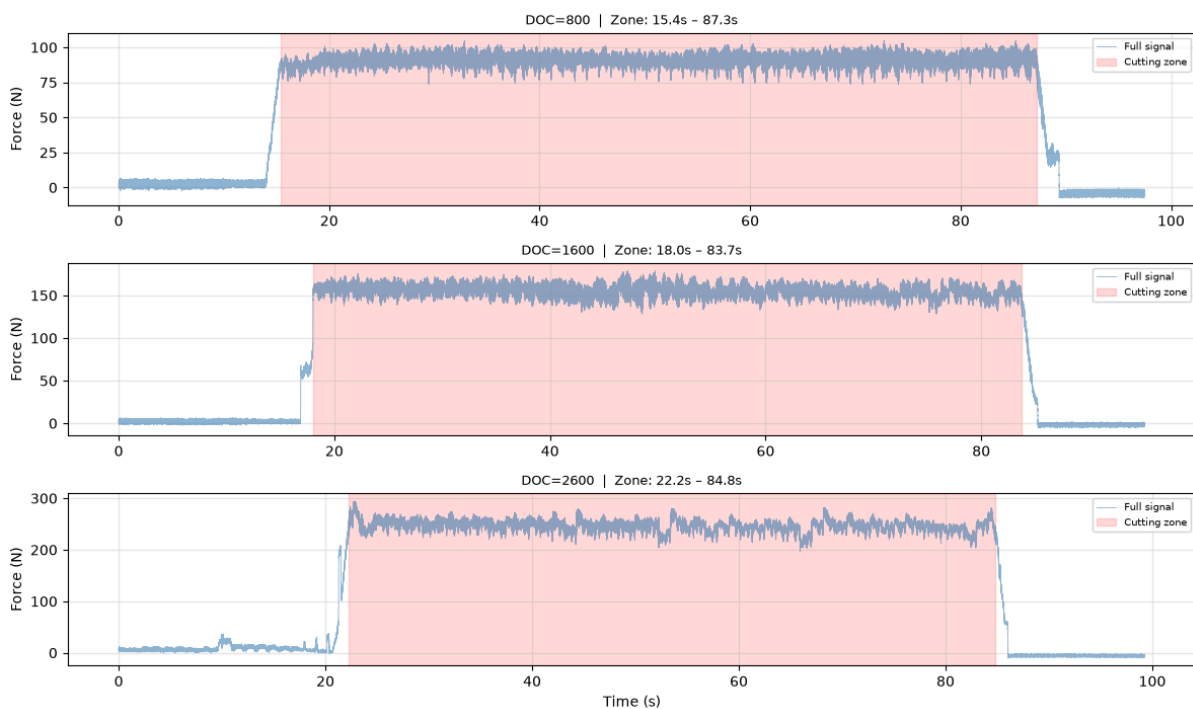
5. Results and Discussions

5.1 At RPM = 52

Cutting Zone Detection:

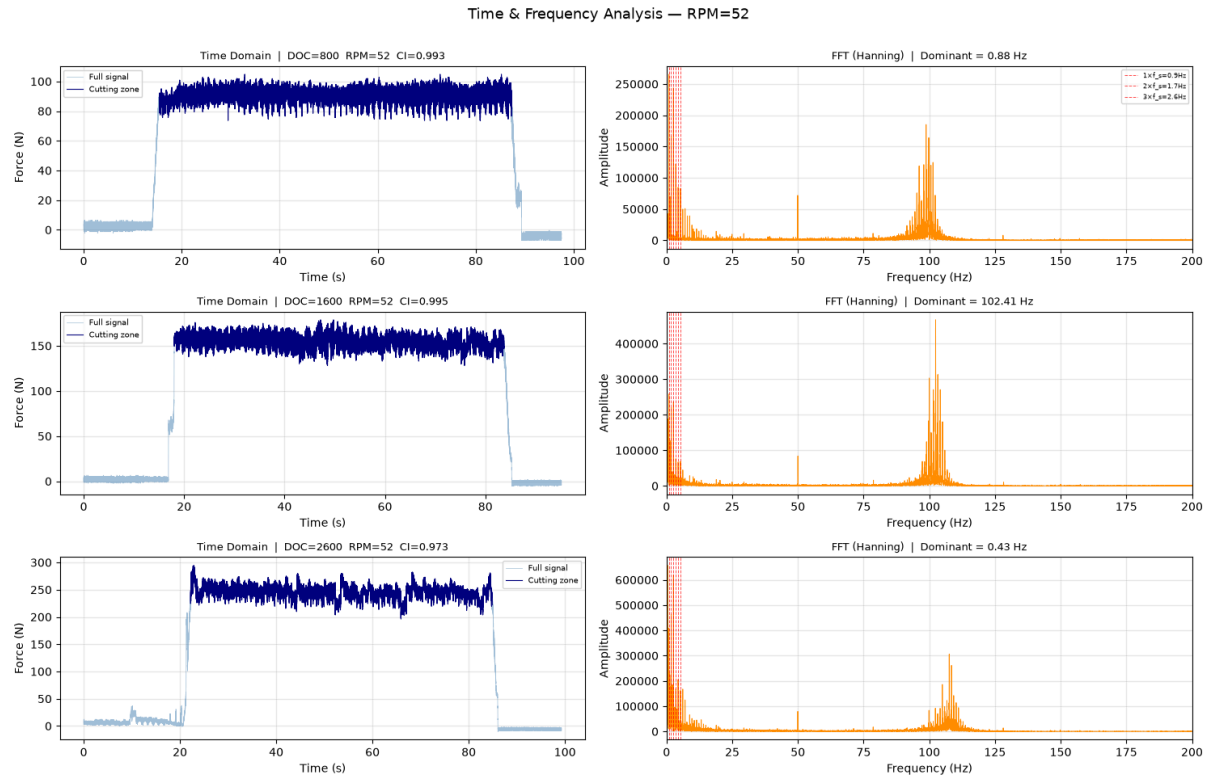
This figure demonstrates automated cutting zone detection from measured force signals at a constant speed of 52 RPM across three increasing depths of cut (DOC). The blue trace plots the real-time cutting force (N) over time (s), while the pink shaded regions highlight the detected steady-state machining zones, successfully isolating active engagement from initial approach and exit air-cutting. As the depth of cut increases from DOC = 800 (15.4s–87.3s) to DOC = 1600 (18.0s–83.7s) and DOC = 2600 (22.2s–84.8s), the mean cutting force scales up proportionally from approximately 85 N to 150 N and 250 N, accompanied by noticeably larger signal fluctuations at higher engagement depths.

Cutting Zone Auto-Detection — RPM=52



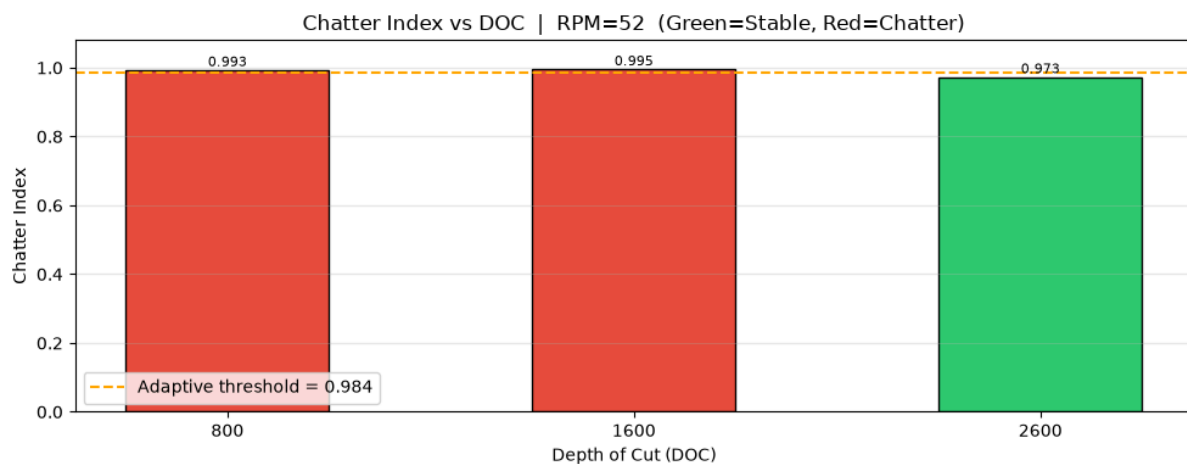
Time and Frequency Analysis:

This figure presents a combined **Time and Frequency Domain Analysis** of cutting forces at a constant spindle speed of **52 RPM** across three depths of cut (DOCs) of 800, 1600, and 2600. The left column displays the time-domain force profiles (N) with high-confidence isolated cutting zones (CI values between 0.973 and 0.995), showing mean forces scaling from ~90 N to ~250 N. The right column displays the corresponding Hanning-windowed FFT spectra (0–200 Hz), highlighting spindle harmonic frequencies ($1 \times f_s = 0.87\text{Hz}$, $2 \times f_s = 1.73\text{Hz}$, $3 \times f_s = 2.6\text{Hz}$) in red dashed lines alongside high-amplitude structural dynamic responses centered around 100 Hz.



Chatter Index vs Depth of Cut:

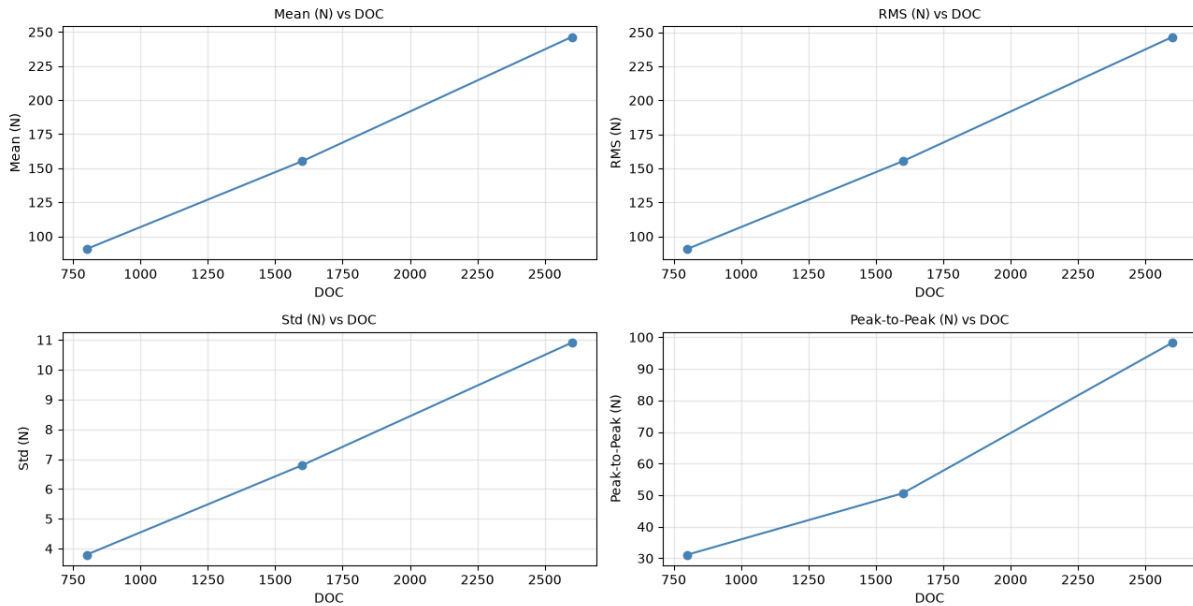
At **RPM = 52**, the chatter index is **0.993**, **0.995**, and **0.973** for DOCs of **800**, **1600**, and **2600 μm**, respectively. The adaptive threshold is **0.984**. Therefore, the 800 μm and 1600 μm conditions lie above the threshold and are classified as **Chatter**, while the 2600 μm condition lies below the threshold and is classified as **Stable**. This indicates that, for this particular RPM, the empirical chatter classification does not show a monotonic variation with DOC.



Force Magnitude Feature Trends:

For RPM = 52, all four force-magnitude parameters show a clear increasing trend with depth of cut. The mean force and RMS force increase from approximately 91 N at 800 μm DOC to 245 N at 2600 μm DOC. Similarly, the standard deviation increases from about 3.8 N to 10.9 N, while the peak-to-peak force increases from approximately 31 N to 98 N.

Force Magnitude Feature Trends — RPM=52

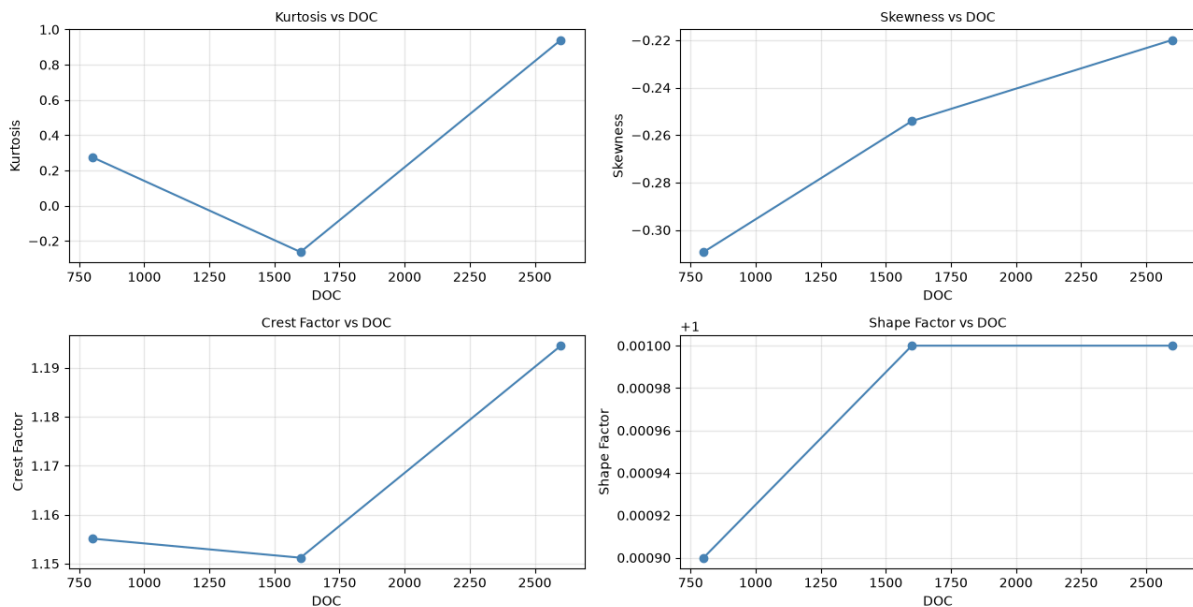


Signal Shape Features:

For RPM = 52, the signal-shape features exhibit different trends as DOC increases. Kurtosis decreases from approximately 0.28 at 800 μm to -0.27 at 1600 μm , before increasing sharply to about 0.93 at 2600 μm , indicating a considerable change in the distribution of force fluctuations at the highest DOC. Skewness remains negative throughout but decreases, changing from approximately -0.31 to -0.22, indicating a gradual shift in the asymmetry of the force distribution.

The **crest factor** remains nearly constant between 800 and 1600 μm and then increases to approximately **1.19 at 2600 μm** , suggesting a greater contribution of peak fluctuations at the highest DOC. The **shape factor** increases slightly from approximately **0.00090 at 800 μm to 0.00100 at 1600 μm** and then remains nearly constant at 2600 μm . Overall, the most pronounced changes in the signal-shape characteristics occur at the highest DOC, particularly in **kurtosis and crest factor**.

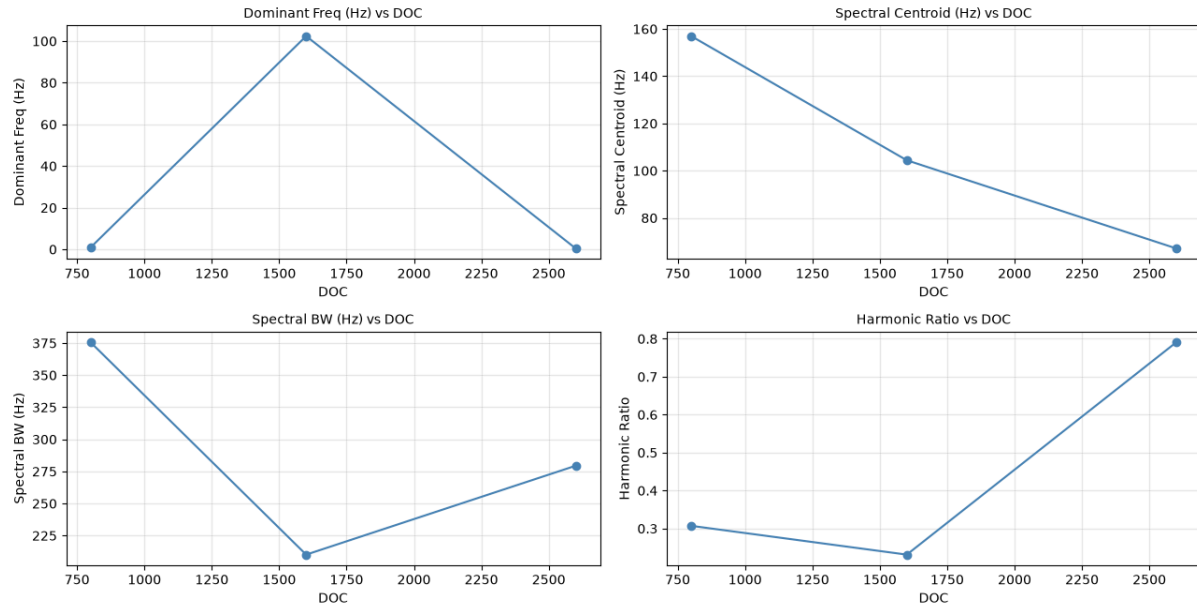
Signal Shape Feature Trends — RPM=52



Spectral Features:

At 52 RPM, the spectral features exhibit a non-monotonic variation with DOC. The dominant frequency reaches approximately 102.41 Hz at 1600 μm , while remaining close to 0 Hz at 800 and 2600 μm . The spectral centroid decreases progressively from approximately 157 Hz to 67 Hz as DOC increases, whereas the spectral bandwidth decreases initially and then increases at the highest DOC. The harmonic ratio decreases from 0.31 at 800 μm to 0.23 at 1600 μm , then increases substantially to 0.79 at 2600 μm , indicating a significant redistribution of spectral energy among the investigated conditions.

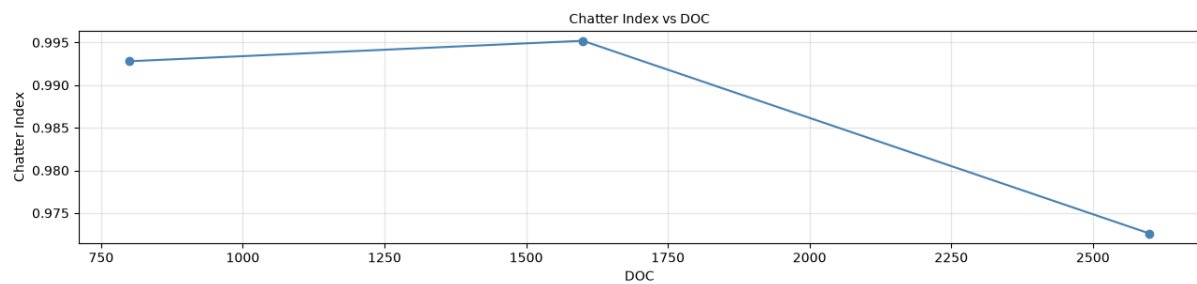
Spectral Feature Trends — RPM=52



Chatter Feature:

The chatter index remains very high across all three DOC conditions. It increases slightly from approximately 0.993 at 800 μm to 0.995 at 1600 μm , before decreasing to approximately 0.973 at 2600 μm . Thus, the chatter index does not show a monotonic relationship with DOC for this spindle speed. The highest value occurs at 1600 μm , while the lowest occurs at 2600 μm . According to the empirical CI-based classification used in this study, the 2600 μm condition therefore exhibits a comparatively lower tendency toward the spectral behavior defined as chatter.

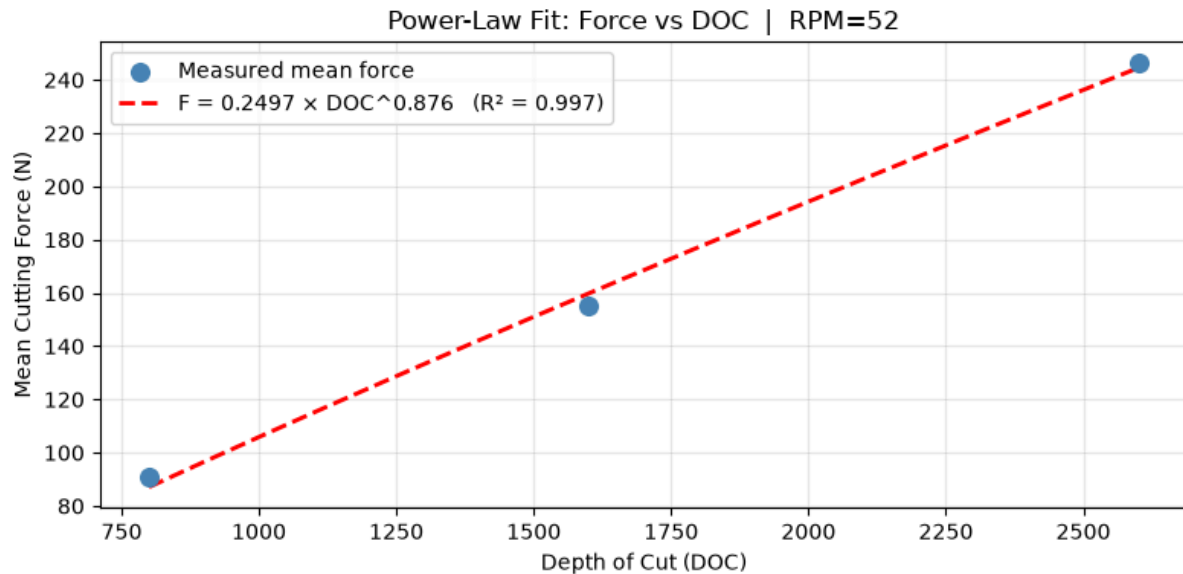
Chatter Feature Trends — RPM=52



Power-Law Fit: Force vs DOC:

For RPM = 52, the mean cutting force shows a strong power-law relationship with DOC. The fitted model is $F = 0.2497 \cdot \text{DOC}^{0.876}$, with a coefficient of determination of $R^2 = 0.997$.

The high R^2 value indicates that the power-law model provides an excellent fit to the measured cutting-force data for this spindle speed. The exponent $n=0.876$ is less than 1, indicating a **sublinear increase** in mean cutting force as DOC increases. Overall, the results demonstrate a strong and consistent dependence of cutting-force magnitude on DOC at 52 RPM.



Pearson Correlation with Depth of Cut:

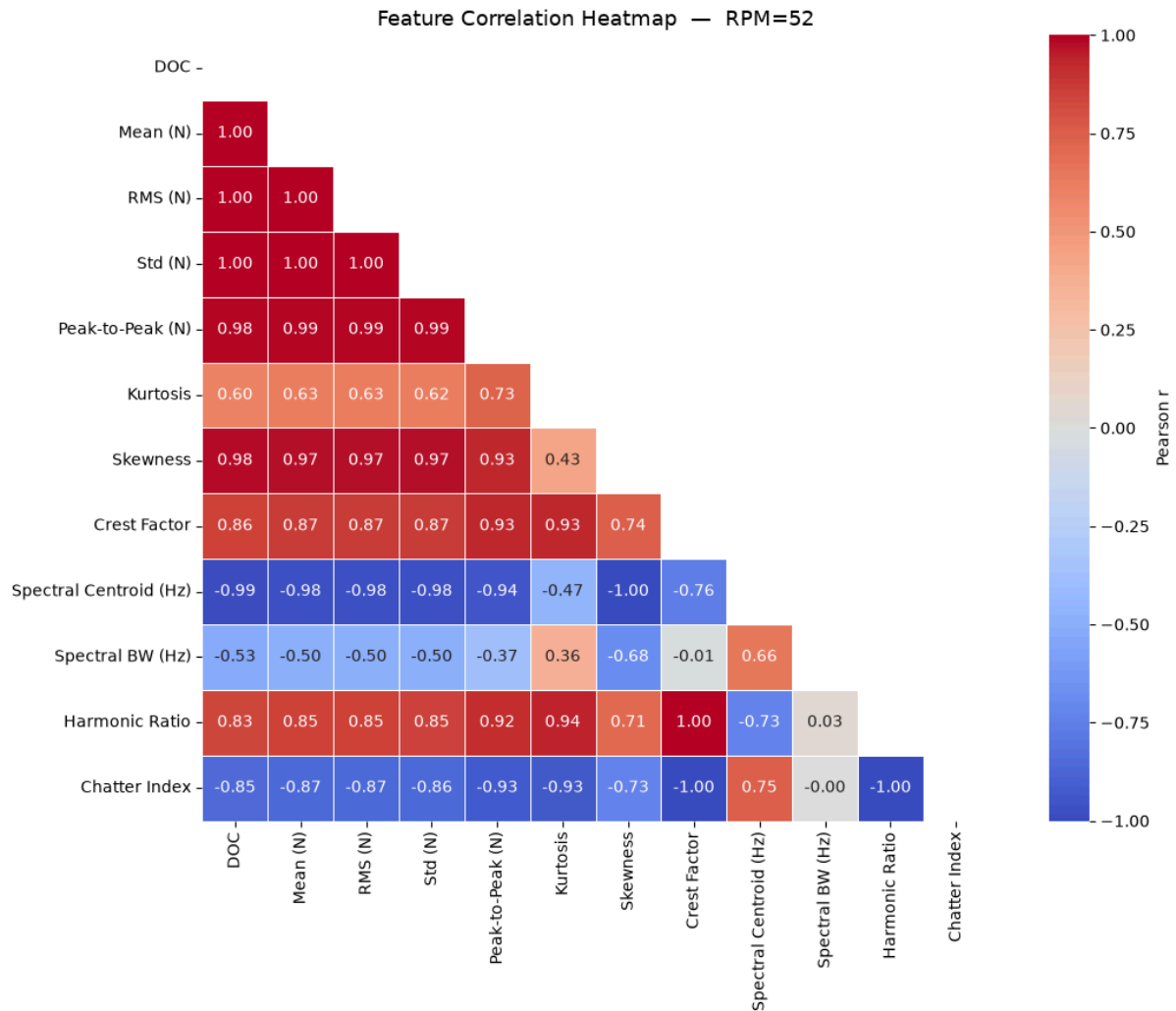
Pearson correlation analysis was performed at 52 RPM to evaluate the relationship between DOC and the extracted time-domain, signal-shape, and frequency-domain features. The correlation coefficient r indicates the strength and direction of the linear relationship, while the corresponding p -value assesses its statistical significance. A relationship is considered statistically significant when $p < 0.05$.

Pearson Correlation with DOC (RPM=52)

Mean (N)	$r = +0.999$	$p = 0.0222$	✓ significant
RMS (N)	$r = +0.999$	$p = 0.0222$	✓ significant
Std (N)	$r = +1.000$	$p = 0.0163$	✓ significant
Peak-to-Peak (N)	$r = +0.985$	$p = 0.1108$	✗ not significant
Kurtosis	$r = +0.604$	$p = 0.5872$	✗ not significant
Skewness	$r = +0.980$	$p = 0.1266$	✗ not significant
Crest Factor	$r = +0.857$	$p = 0.3444$	✗ not significant
Shape Factor	$r = +0.832$	$p = 0.3741$	✗ not significant
Impulse Factor	$r = +0.858$	$p = 0.3429$	✗ not significant
Dominant Freq (Hz)	$r = -0.068$	$p = 0.9568$	✗ not significant
Spectral Centroid (Hz)	$r = -0.987$	$p = 0.1029$	✗ not significant
Spectral BW (Hz)	$r = -0.526$	$p = 0.6477$	✗ not significant
Harmonic Ratio	$r = +0.834$	$p = 0.3723$	✗ not significant
Chatter Index	$r = -0.849$	$p = 0.3542$	✗ not significant

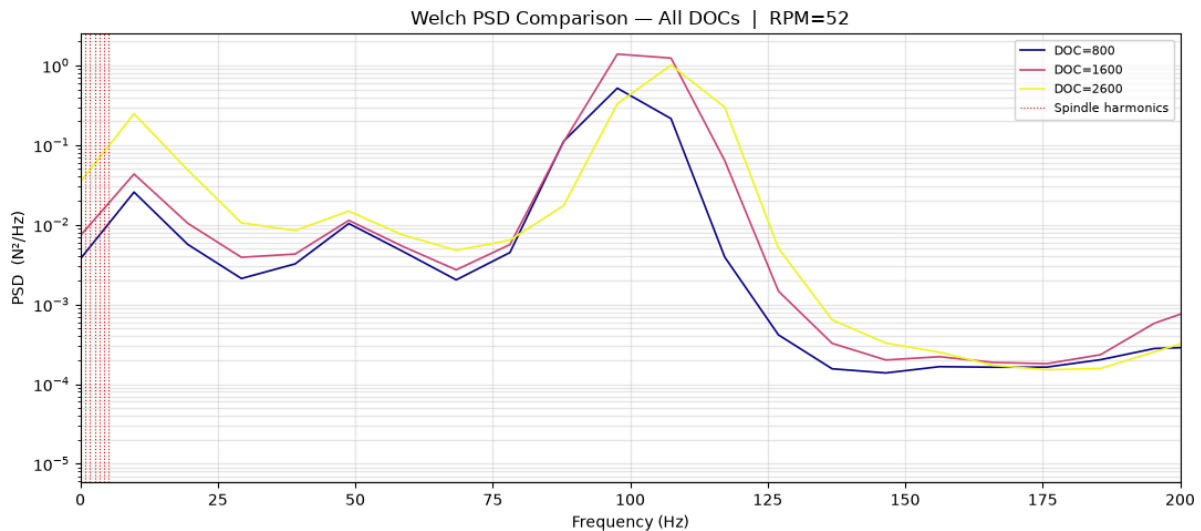
Feature Correlation Heatmap:

The heatmap shows the relationships among DOC and the extracted features. Mean force, RMS, and standard deviation show very strong positive correlations with DOC, indicating similar trends. Spectral centroid and chatter index show strong negative correlations with several force-related features, while spectral bandwidth shows comparatively weaker correlations.



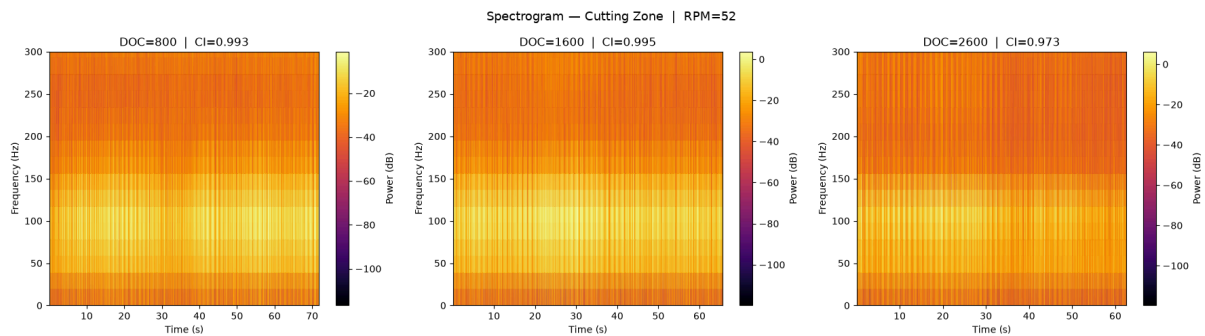
Welch PSD Comparison:

The figure shows a Welch PSD comparison across three depths of cut (DOC = 800, 1600, and 2600 μm) at a spindle speed of 52 RPM. All three spectra follow a similar overall shape: low and relatively flat energy below 40 Hz, a broad low-amplitude plateau between 40–90 Hz, a single dominant peak in the 90–115 Hz range, and a steep roll-off beyond 120 Hz down to a noise floor near 200 Hz. The dominant peak is highest for DOC = 1600 μm ($\sim 1.4 \text{ N}^2/\text{Hz}$), followed by DOC = 2600 μm ($\sim 1.1 \text{ N}^2/\text{Hz}$), and the lowest for DOC = 800 μm ($\sim 0.5 \text{ N}^2/\text{Hz}$). The peak location also shifts slightly: the DOC = 800 μm and DOC = 1600 μm peaks are near 97 Hz, while the DOC = 2600 μm peak is further to the right, near 108 Hz. The spindle harmonics, marked by the red dotted lines, are confined to frequencies below approximately 5 Hz. At the same time, the dominant spectral peak for all three conditions lies well outside this range, showing that spectral energy at this spindle speed is concentrated away from the spindle-related frequencies rather than around them.



Spectrogram Analysis:

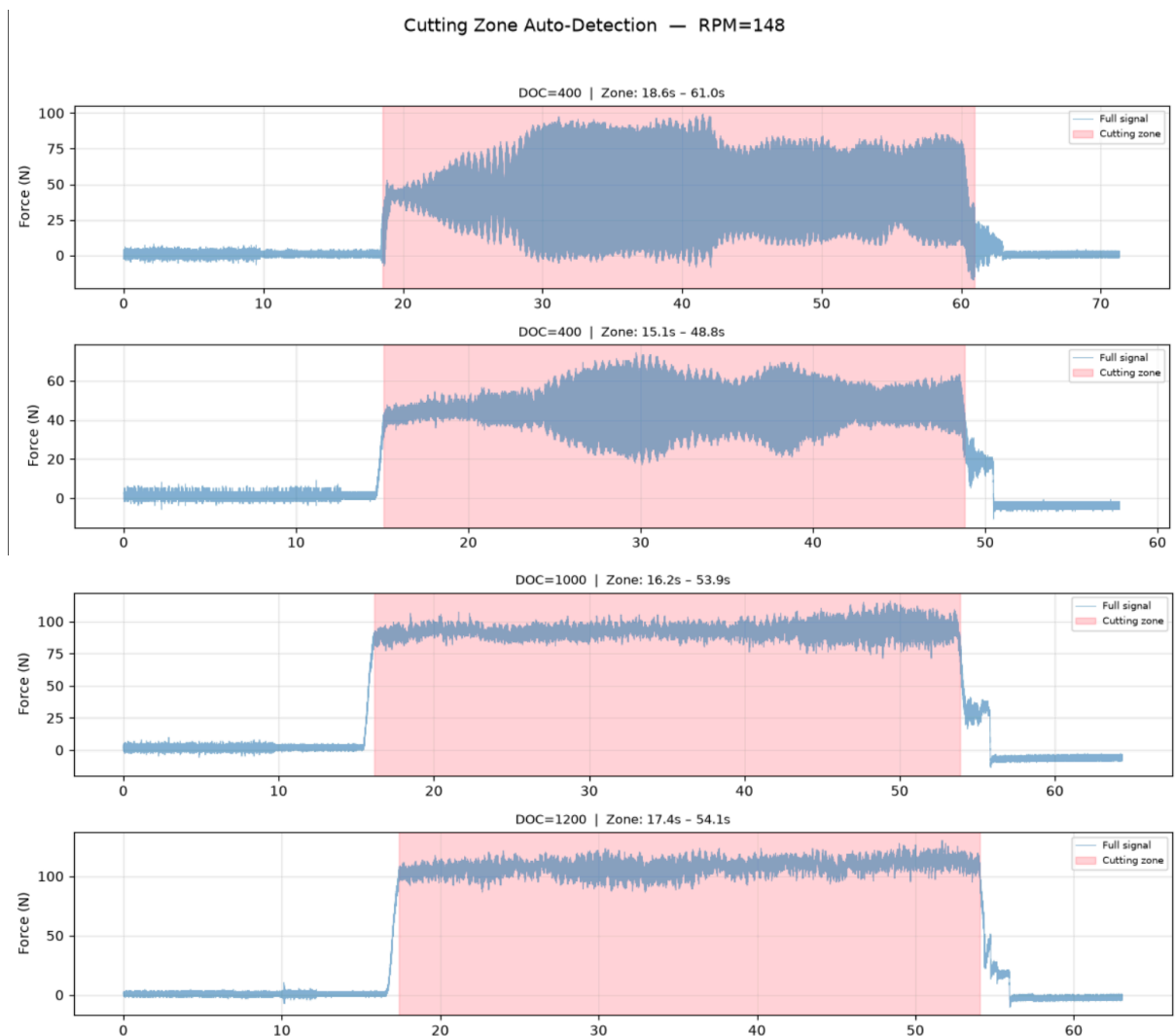
The stationarity seen here matters for the chatter assessment: since the dominant 50–150 Hz band stays fixed in position and strength throughout the cut for all three DOCs, there's no sign of the frequency content drifting, broadening, or intensifying over time — which is what you'd expect to see if chatter were building up progressively during the cut (e.g., regenerative chatter growing as vibration marks compound pass-over-pass). This temporal stability supports treating the process as being in a steady dynamic state throughout the cutting zone, meaning the single time-averaged Chatter Index computed from the FFT/PSD is a fair representative measure for the whole cut, not just a snapshot that misses instability building up partway through. In other words, the spectrogram's main contribution here isn't to reveal a new chatter signature, but to confirm that the time-averaged CI value is missing a time-localized transition into chatter reported for each condition.

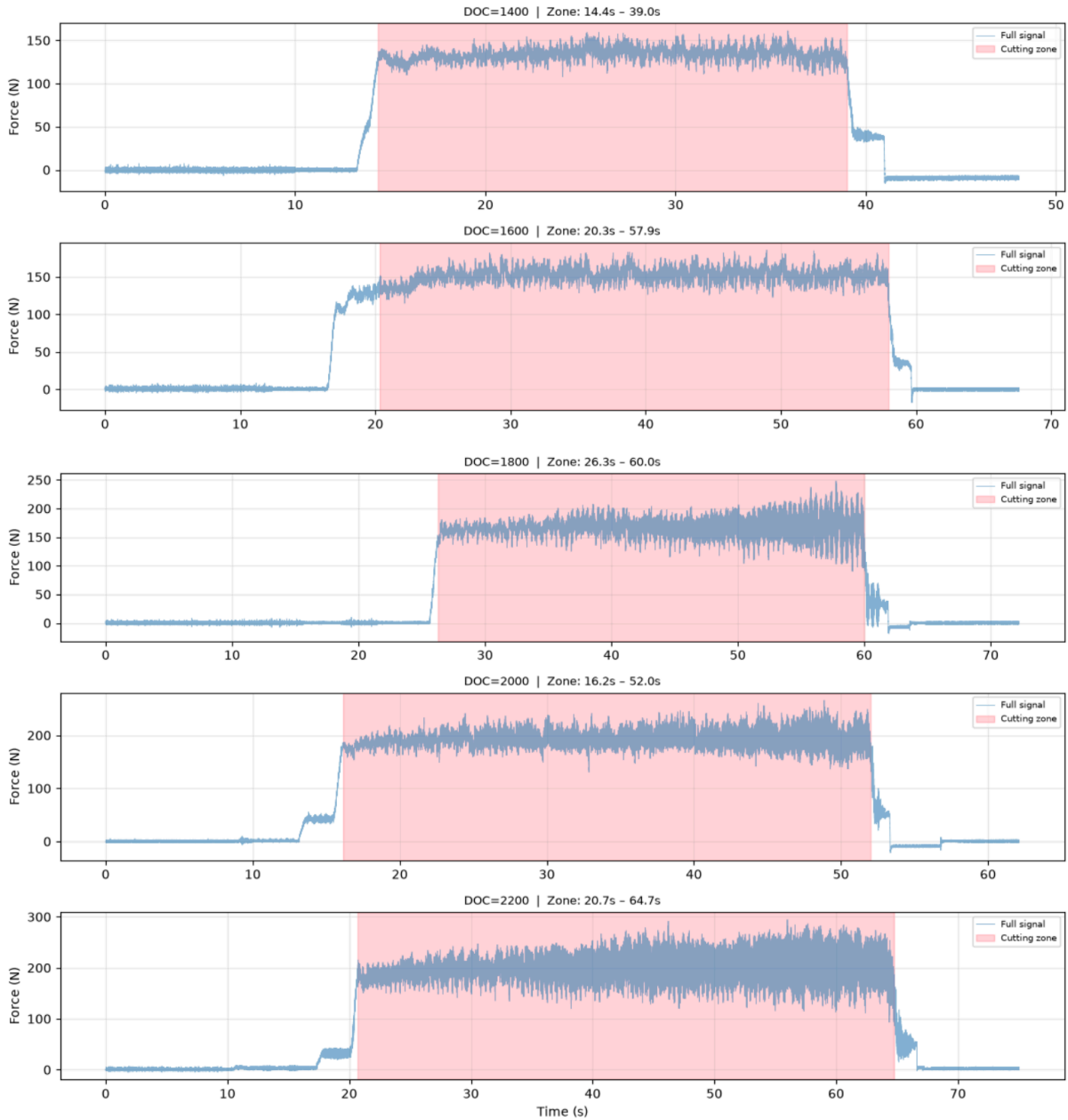


5.2 At RPM = 148

Cutting Zone Detection:

At 148 RPM, the amplitude-based cutting-zone detection method successfully identifies the active machining regions for all investigated depths of cut. The detected zones begin after the initial idle portion of the signal and terminate before the tool-retraction region, effectively separating the steady-state cutting phase from non-cutting regions. The detected cutting intervals vary with DOC, ranging approximately from **14.4–39.0 s for DOC = 1400 μm** to **20.7–64.7 s for DOC = 2200 μm** . A clear increase in cutting force magnitude is observed as DOC increases. At lower DOC values, such as **400–1000 μm** , the force remains comparatively lower, whereas at **1800–2200 μm** the force reaches substantially higher levels and exhibits stronger fluctuations. The higher-DOC signals also exhibit greater variation in force amplitude throughout the cutting interval, indicating a higher machining load and more pronounced dynamic behavior. Overall, the automatic detection provides consistent cutting intervals across the different machining conditions, ensuring that the subsequent statistical, FFT, PSD, and chatter analyses are performed on the actual material-removal region rather than on idle or transient portions of the recorded signal.

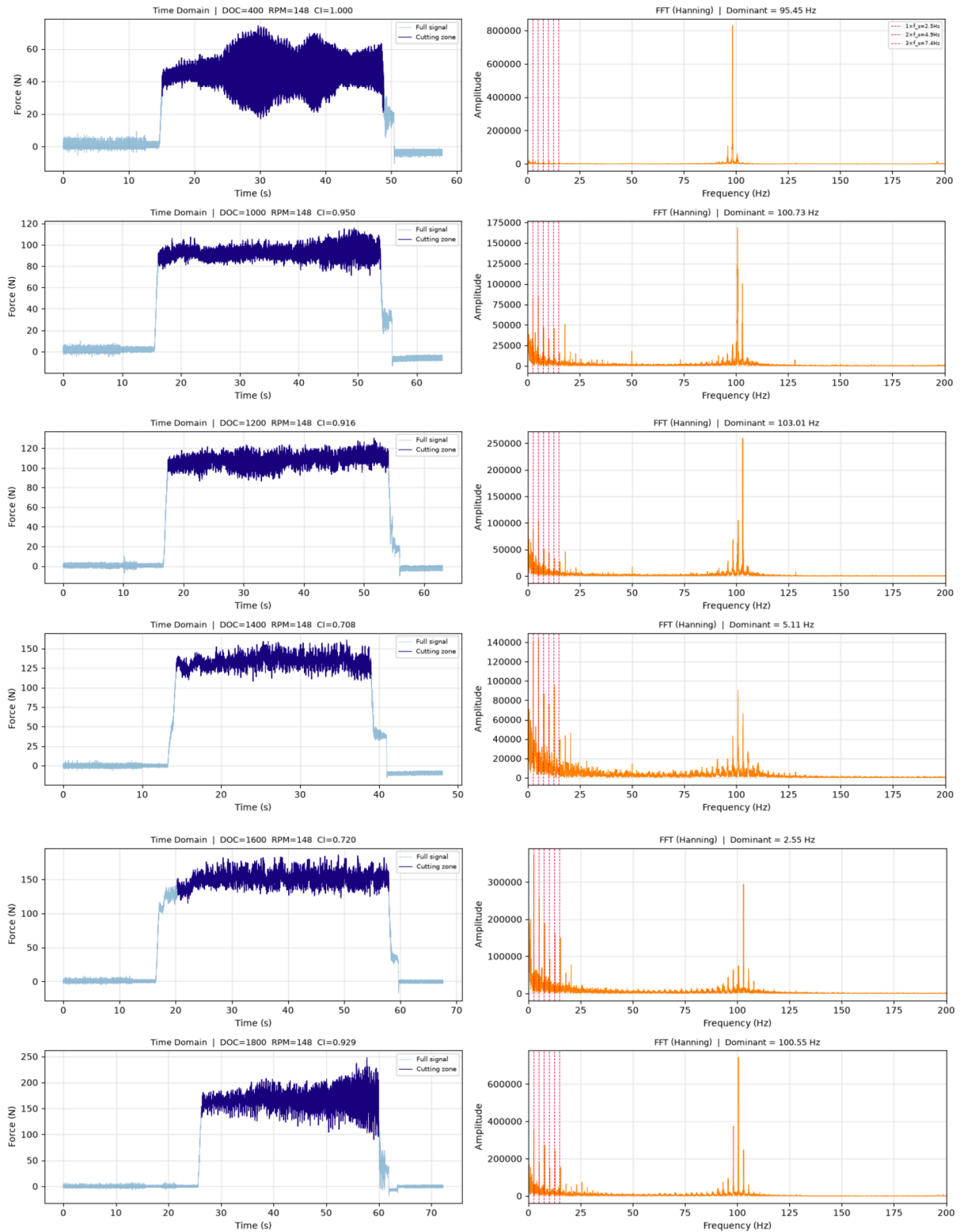


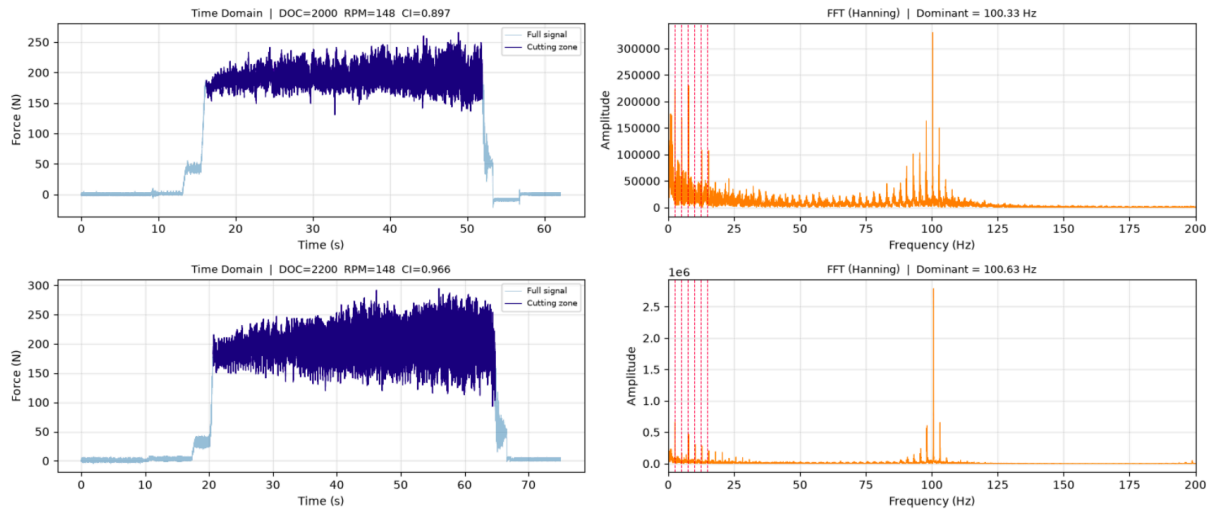


Time and Frequency Analysis:

At 148 RPM, the cutting force increases progressively with DOC, from approximately 40–60 N at 400 μm to 200–250 N at 2200 μm , with greater fluctuations at higher DOC. The FFT spectra show dominant frequencies primarily around 100 Hz, except at 1400 and 1600 μm , where they shift to approximately 5.11 Hz and 2.55 Hz, respectively. Overall, increasing DOC significantly affects both the force magnitude and frequency characteristics of the machining process.

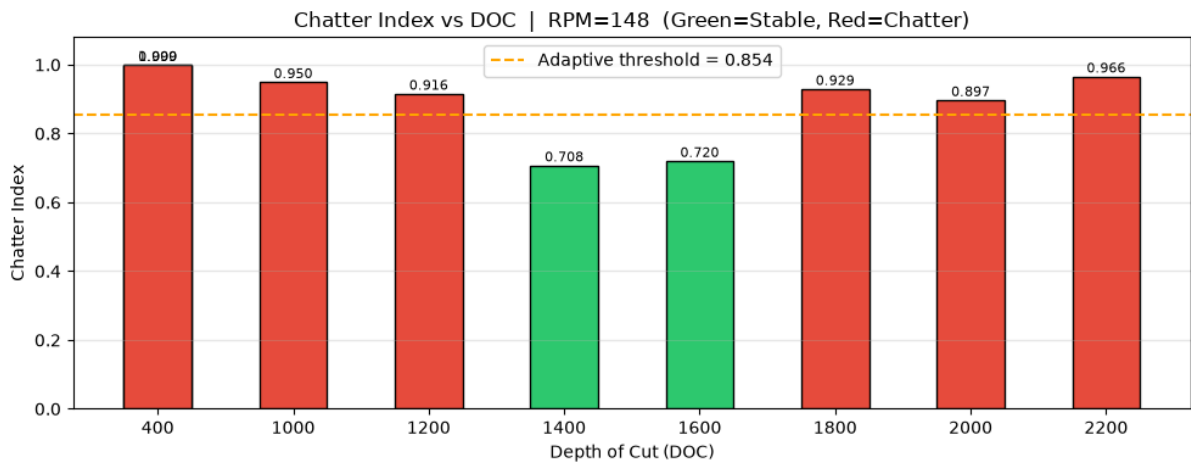
Time & Frequency Analysis — RPM=148





Chatter Index vs Depth of Cut:

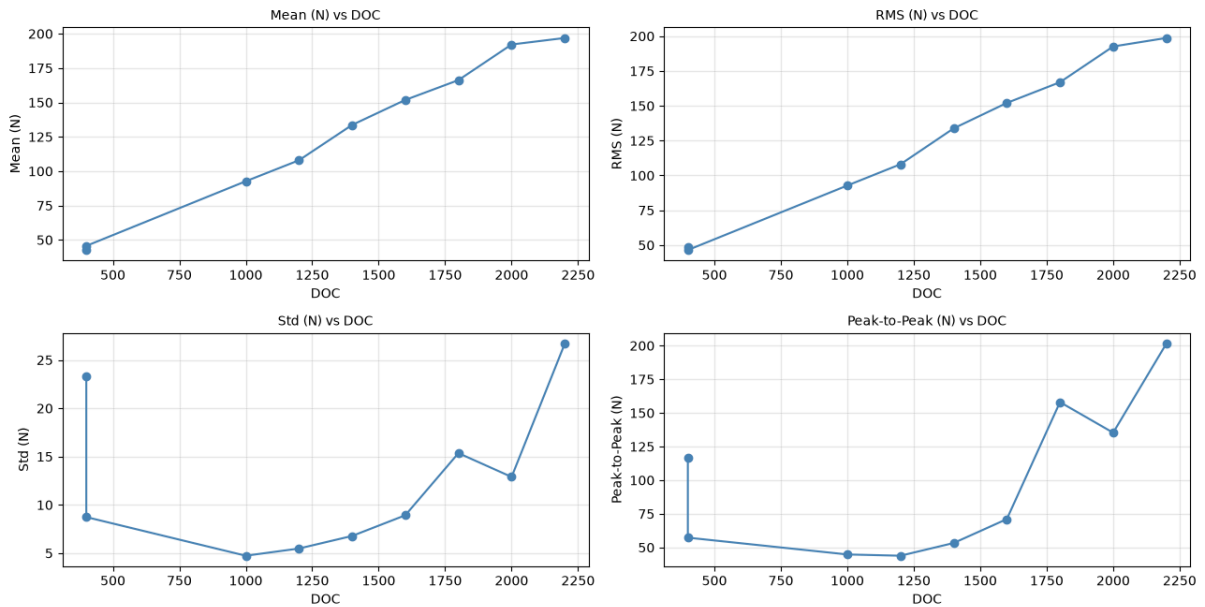
At 148 RPM, the adaptive chatter-index threshold is 0.854. The conditions at 1400 μm (CI = 0.708) and 1600 μm (CI = 0.720) are classified as Stable, while all other DOC conditions are classified as Chatter. The highest chatter index occurs at 400 μm (CI = 1.000), whereas the lowest occurs at 1400 μm (CI = 0.708). Overall, the chatter behavior shows a non-monotonic relationship with DOC, with a temporary stable region at 1400–1600 μm .



Force Magnitude Feature Trends:

At 148 RPM, the mean force and RMS increase consistently with DOC, rising from approximately 45 N at 400 μm to nearly 200 N at 2200 μm . The standard deviation shows a generally increasing trend at higher DOC, reaching its maximum at 2200 μm , indicating greater force fluctuations. Similarly, the peak-to-peak force increases substantially at higher DOC, reaching approximately 200 N at 2200 μm . Overall, increasing DOC clearly increases both the magnitude and variability of cutting force.

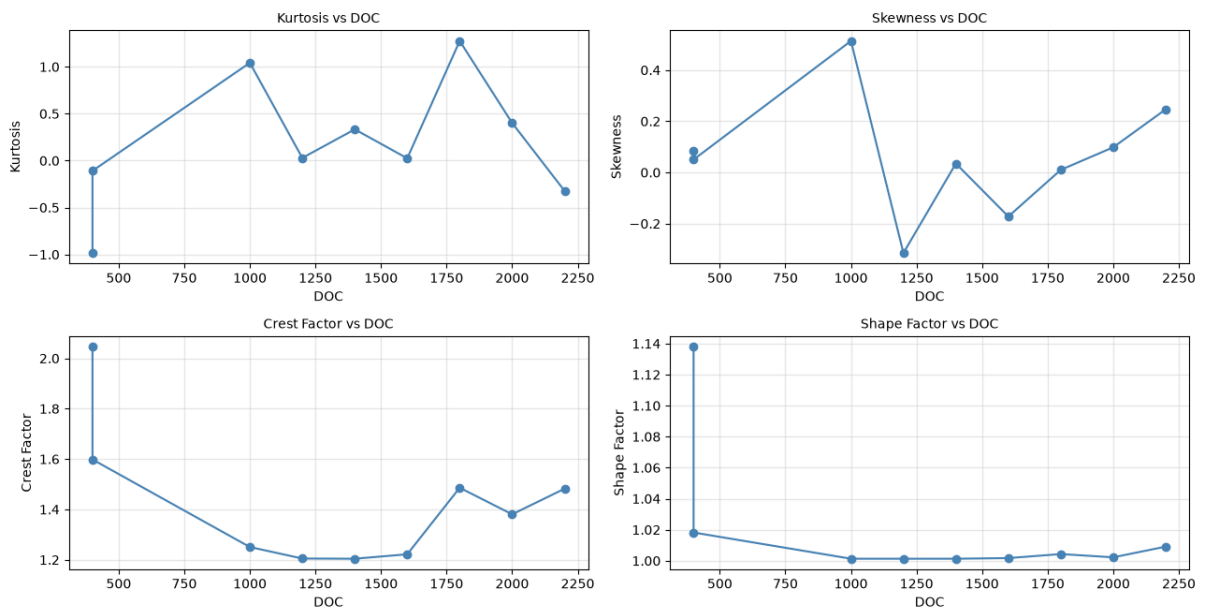
Force Magnitude Feature Trends — RPM=148



Signal Shape Features:

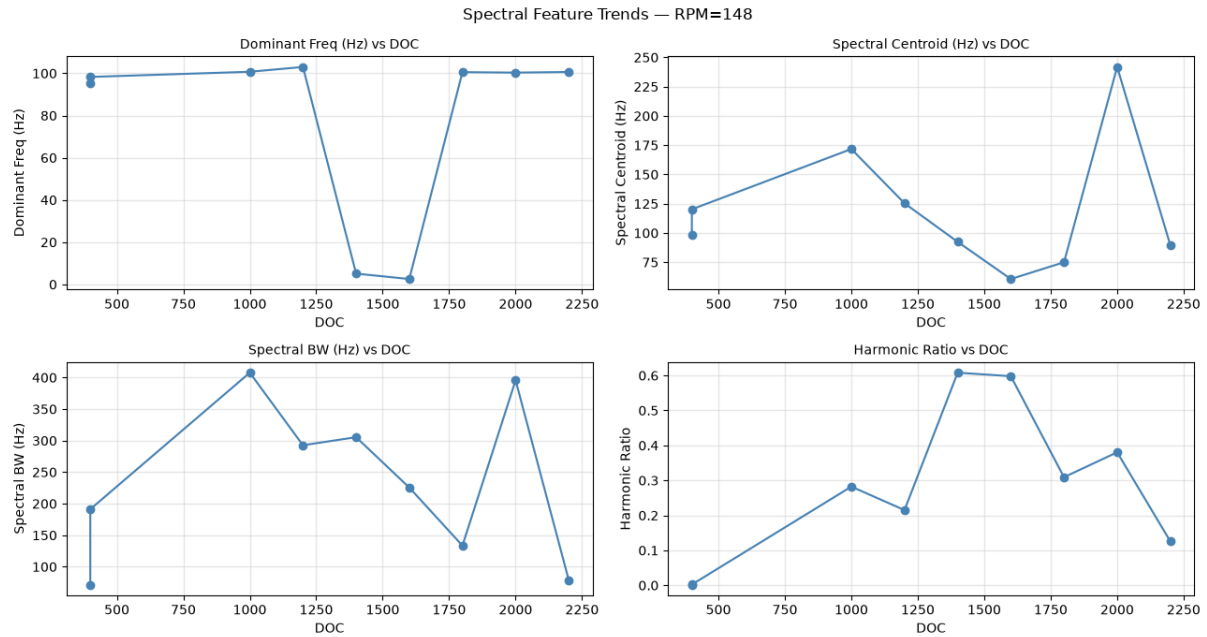
At a constant RPM of 148, the signal shape features exhibit distinct non-linear trends across varying Depths of Cut (DOC). At a low DOC (~ 400), Crest Factor and Shape Factor reach their absolute maximums before sharply declining, with Shape Factor flattening near 1.00 and Crest Factor dipping to a low around DOC 1200–1600 before slightly recovering. Meanwhile, Kurtosis and Skewness display significant fluctuations throughout the range: Skewness peaks near DOC 1000 before dropping to its minimum around DOC 1200 and gradually rising, whereas Kurtosis fluctuates heavily, reaching its peak near DOC 1800 before falling off at higher DOC levels.

Signal Shape Feature Trends — RPM=148



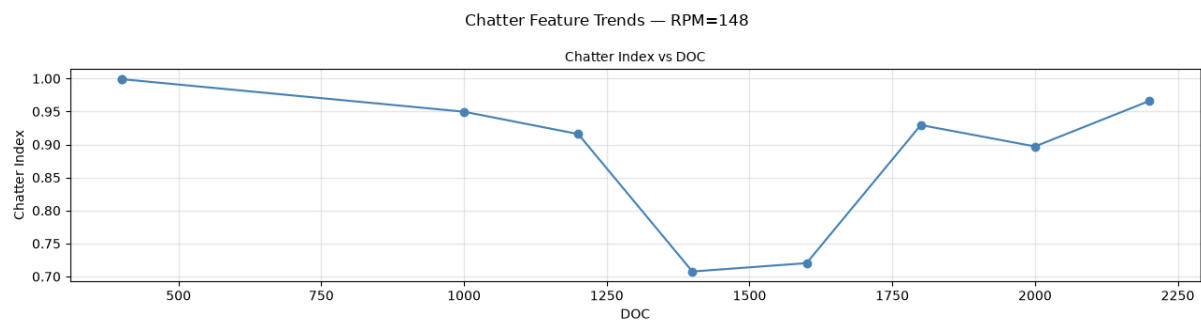
Spectral Features:

At a constant RPM of 148, the spectral feature trends across varying Depth of Cut (DOC) values reveal notable structural shifts in the frequency domain. Dominant Frequency stays high ($\sim 100\text{Hz}$) for low-to-mid DOCs, sharply plummets to near 0Hz between DOC 1400 and 1600, and then recovers back to 100Hz at higher DOCs. Spectral Centroid stays mostly under 175Hz across lower DOCs, except for a sharp, isolated peak above 240Hz at DOC 2000. Spectral Bandwidth (BW) fluctuates significantly, showing dual prominent peaks around DOC 1000 ($\sim 410\text{Hz}$) and DOC 2000 ($\sim 400\text{Hz}$). Finally, Harmonic Ratio steadily increases from 0 at a low DOC, reaching a high plateau 0.6 between DOC 1400 and 1600 corresponding directly to the drop in dominant frequency, before declining again at higher DOC levels.



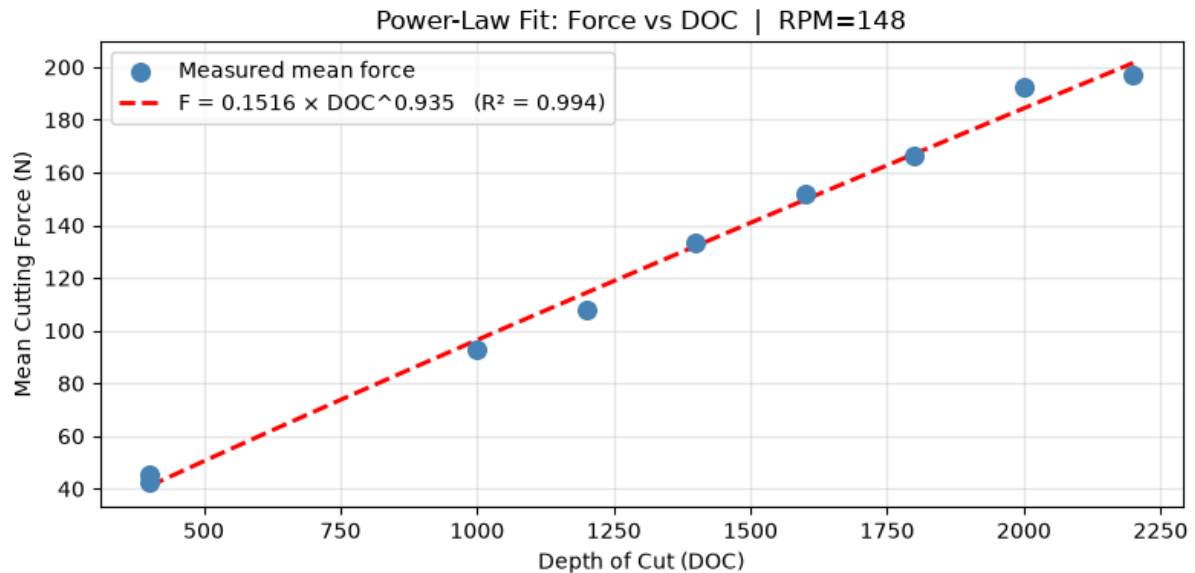
Chatter Feature:

At RPM = 148, the Chatter Index exhibits a clear V-shaped variation with DOC. It starts at its maximum value of 1.000 at DOC = 400 μm and gradually decreases to approximately 0.92 at DOC = 1200 μm . It then drops sharply to a minimum of approximately 0.71–0.72 between DOC = 1400 and 1600 μm — the same region where the dominant frequency collapses to near 0 Hz and the harmonic ratio peaks. Following this dip, the index rebounds to approximately 0.93 at DOC = 1800 μm , then continues rising to around 0.97 at DOC = 2200 μm .



Power-Law Fit: Force vs DOC:

At RPM = 148, the mean cutting force again shows a strong power-law relationship with DOC. The fitted model is $F = 0.1516(\text{DOC})^{0.935}$, with a coefficient of determination (R^2) = 0.994. As at RPM = 52, the exponent $n = 0.935$ is less than 1, indicating a sub-linear increase in mean cutting force with depth of cut — though this exponent is closer to 1 (nearer-linear) than at RPM = 52 ($n = 0.876$), suggesting a slightly different force-scaling behavior at this spindle speed.



Pearson Correlation with Depth of Cut:

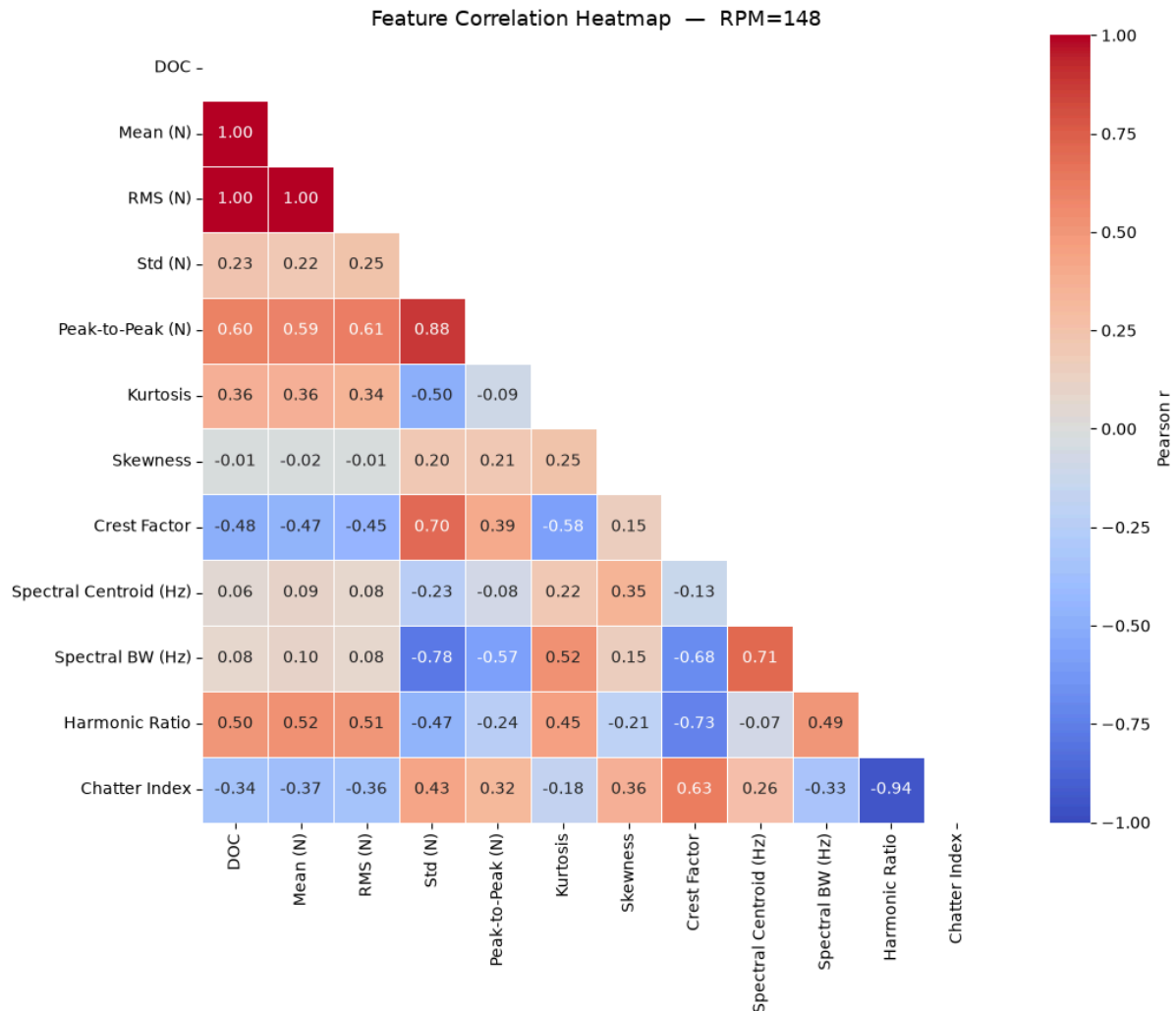
At a constant RPM of 148, the Pearson correlation analysis reveals that only Mean (N) and RMS (N) share statistically significant, near-perfect positive linear relationships with Depth of Cut (DOC), both yielding $r = +0.997$ ($p = 0.0000$). In contrast, all higher-order statistical, shape, spectral, and chatter metrics fail to reach statistical significance. Among the non-significant features, Peak-to-Peak force ($r = +0.599$, $p = 0.0882$) and Shape Factor ($r = -0.580$, $p = 0.1015$) exhibit moderate linear trends. In contrast, metrics such as Skewness ($r = -0.011$, $p = 0.9781$) and Spectral Centroid ($r = +0.062$, $p = 0.8734$) show virtually no linear correlation with DOC.

Pearson Correlation with DOC (RPM=148)

Mean (N)	$r = +0.997$	$p = 0.0000$	✓ significant
RMS (N)	$r = +0.997$	$p = 0.0000$	✓ significant
Std (N)	$r = +0.232$	$p = 0.5485$	✗ not significant
Peak-to-Peak (N)	$r = +0.599$	$p = 0.0882$	✗ not significant
Kurtosis	$r = +0.360$	$p = 0.3406$	✗ not significant
Skewness	$r = -0.011$	$p = 0.9781$	✗ not significant
Crest Factor	$r = -0.478$	$p = 0.1935$	✗ not significant
Shape Factor	$r = -0.580$	$p = 0.1015$	✗ not significant
Impulse Factor	$r = -0.513$	$p = 0.1580$	✗ not significant
Dominant Freq (Hz)	$r = -0.119$	$p = 0.7611$	✗ not significant
Spectral Centroid (Hz)	$r = +0.062$	$p = 0.8734$	✗ not significant
Spectral BW (Hz)	$r = +0.084$	$p = 0.8304$	✗ not significant
Harmonic Ratio	$r = +0.499$	$p = 0.1712$	✗ not significant
Chatter Index	$r = -0.343$	$p = 0.3658$	✗ not significant

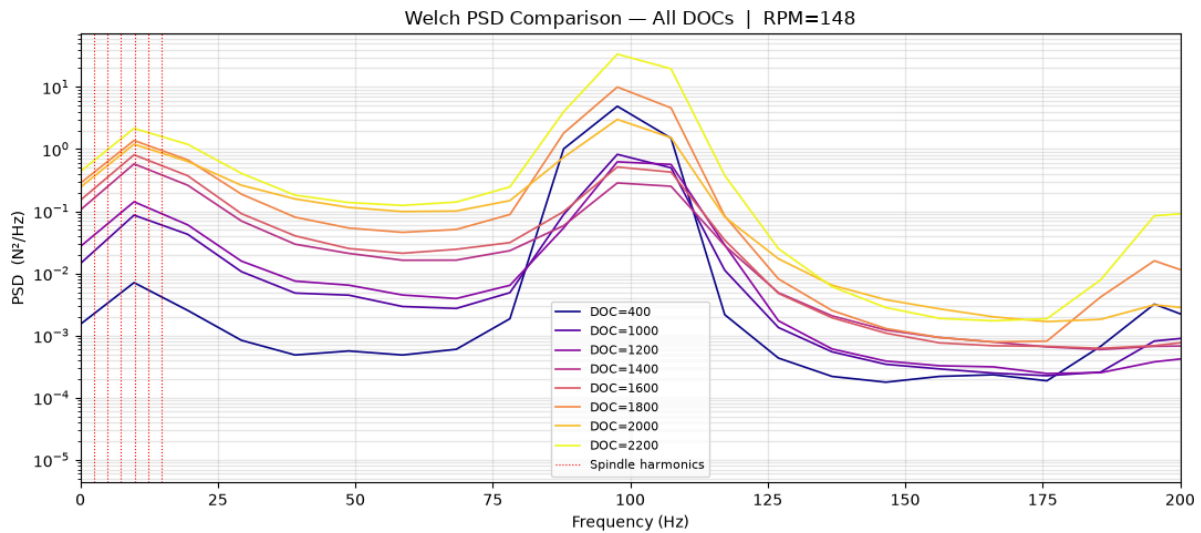
Feature Correlation Heatmap:

At a constant RPM of 148, the feature correlation heatmap highlights strong inter-feature dependencies across time and frequency domains. Mean (N) and RMS (N) are perfectly collinear and show a near-perfect correlation with DOC ($r = 1.00$). Notable strong relationships include a sharp negative correlation between Chatter Index and Harmonic Ratio ($r = -0.94$), an inverse link between Spectral BW and Std (N) ($r = -0.78$), and strong positive associations between Peak-to-Peak (N) and Std (N) ($r = 0.88$) as well as Spectral Centroid and Spectral BW ($r = 0.71$). Conversely, metrics such as Skewness and Spectral Centroid remain almost completely unaligned with process-scale variations such as DOC, Mean (N), and RMS (N).



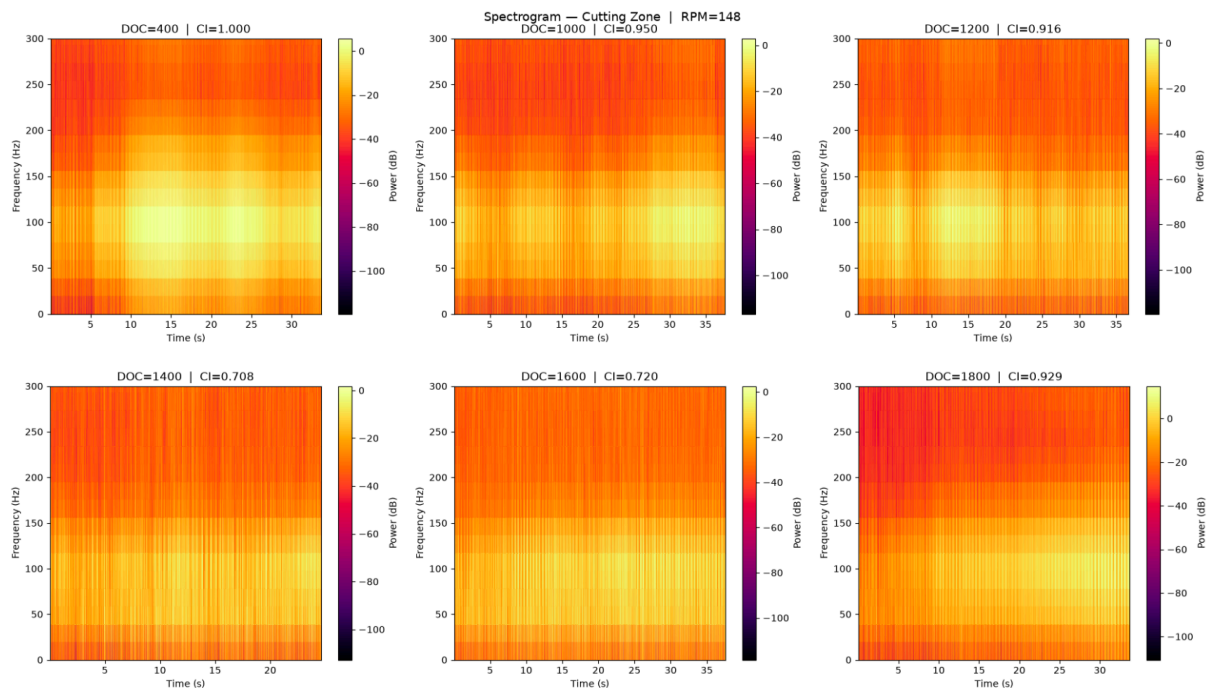
Welch PSD Comparison

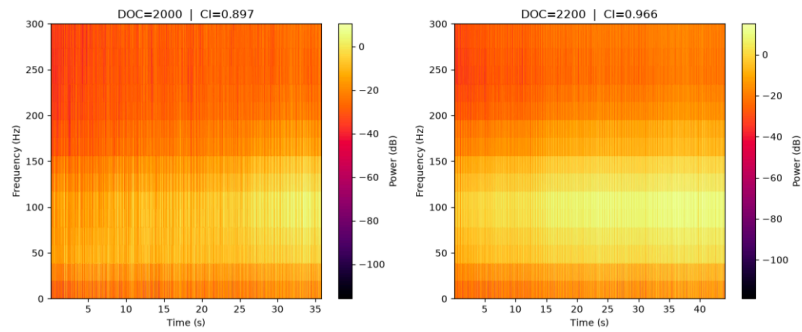
At a constant RPM of 148, the Welch Power Spectral Density (PSD) across all Depths of Cut (DOC) reveals a strong concentration of signal power within specific frequency bands. A secondary peak aligns with low-frequency Spindle harmonics (dotted vertical red lines) below 20Hz, while the dominant power peak spans between 90Hz and 115Hz across all conditions. Overall spectral power scales directly with increasing DOC, with higher DOC values (such as $DOC = 2200$ those shown in yellow) exhibiting PSD magnitudes that are orders of magnitude higher than those at lower DOC values (such as $DOC = 400$ those shown in dark blue). Notable exceptions occur near the dominant peak, where intermediate DOC levels ($DOC = 1400$ and 1600 , shown in magenta/pink) exhibit lower peak energy than surrounding levels, matching the previously observed frequency shifts and chatter dips.



Spectrogram Analysis

At a constant RPM of 148, the time-frequency Spectrograms across varying Depths of Cut (DOC) illustrate steady dynamic energy distribution throughout the cutting duration. For most DOC values ($DOC = 400, 1000, 1200, 1800, 2000, 2200$), signal power remains strongly concentrated in a horizontal band centered around 50–120Hz with high Chatter Index (CI) values ranging from 0.897 to 1.000. However, under intermediate DOC conditions ($DOC = 1400$ with $CI = 0.708$ and $DOC = 1600$ without $CI = 0.720$), this dominant power band noticeably broadens across the spectrum, accompanied by lower overall intensity across all frequencies.





Chatter vs Stable

After running the program on all datasets, the following results were obtained:

RPM ↑	DOC ↓	Chatter Index ↓	State ↓
52	800	0.9917	CHATTER
52	1600	0.9842	CHATTER
52	2600	0.9456	STABLE
114	200	0.9640	CHATTER
114	400	0.9987	CHATTER
114	600	0.9975	CHATTER
114	800	0.9977	CHATTER
114	1000	0.9038	CHATTER
114	1200	0.9706	CHATTER
114	1400	0.9395	CHATTER
114	1800	0.9768	CHATTER
114	2200	0.7818	STABLE

RPM ↑	DOC ↓	Chatter Index ↓	State ↓
148	200	0.9769	CHATTER
148	400	0.9995	CHATTER
148	400	0.9979	CHATTER
148	600	0.9141	CHATTER
148	800	0.8997	CHATTER
148	1000	0.9386	CHATTER
148	1200	0.9004	CHATTER
148	1400	0.7080	STABLE
148	1600	0.7164	STABLE
148	1800	0.9277	CHATTER
148	2000	0.8921	CHATTER
148	2200	0.9649	CHATTER

RPM ↑	DOC ↓	Chatter Index ↓	State ↓
192	200	0.9671	CHATTER
192	200	0.9587	CHATTER
192	400	0.9697	CHATTER
192	400	0.9832	CHATTER
192	600	0.9992	CHATTER
192	800	0.9996	CHATTER
192	800	0.8458	STABLE
192	800	0.7619	STABLE
192	800	0.9658	CHATTER
192	1000	0.9996	CHATTER
192	1000	0.9994	CHATTER
192	1200	0.9996	CHATTER

RPM ↑	DOC ↓	Chatter Index ↓	State ↓
250	200	0.9775	STABLE
250	400	0.9807	STABLE
250	600	0.9997	CHATTER
250	600	0.9997	CHATTER
250	800	0.9998	CHATTER
250	1000	0.9996	CHATTER
325	200	0.9790	CHATTER
325	400	0.9194	STABLE
325	600	0.9989	CHATTER
325	700	0.9996	CHATTER
325	800	0.9997	CHATTER
420	100	0.9986	CHATTER

RPM ↑	DOC ↓	Chatter Index ↓	State ↓
420	200	0.9995	CHATTER
420	400	0.9913	STABLE
420	400	0.9926	STABLE
420	600	0.9998	CHATTER
420	700	0.9950	STABLE
420	800	0.9999	CHATTER
420	900	0.9999	CHATTER
420	1200	1.0000	CHATTER
420	1400	1.0000	CHATTER
550	200	0.9559	STABLE
550	400	0.9918	CHATTER
550	600	0.9967	CHATTER